

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Kyle tried to use the table of integrals to evaluate the integral $\int (5-4x^2)^{3/2} dx$. His work is shown below. His final answer is not correct. What did he do wrong?

Kyle's Work:

We use entry #92 from the table of integrals in our textbook which states that

$$\int (a^2 - u^2)^{3/2} du = -\frac{u}{8}(2u^2 - 5a^2)\sqrt{a^2 - u^2} + \frac{3a^4}{8}\sin^{-1}\left(\frac{u}{a}\right) + C \quad \text{if } a > 0.$$

Clearly here $a = \sqrt{5}$ & $u = 2x$, so this means that

$$\int (5-4x^2)^{3/2} dx = -\frac{2x}{8}\left(2(2x)^2 - 5(\sqrt{5})^2\right)\sqrt{5-4x^2} + \frac{3(\sqrt{5})^4}{8}\sin^{-1}\left(\frac{2x}{\sqrt{5}}\right) + C$$

2. Use an entry from the table of integrals to evaluate the integral $\int \frac{dx}{\sqrt{2x^3 + 3x^2}}$, for $x > 0$.

3. Evaluate the integral $\int_1^e \frac{1}{x\sqrt{4+\ln^2(x)}} dx$ using an entry from the table of integrals.

4. Entry #93b in the table of integrals from our textbook states that $\int \frac{du}{a^2-u^2} = \frac{1}{2a} \ln \left| \frac{u+a}{u-a} \right| + C$. Prove that this is correct by fully integrating and simplifying $\int \frac{du}{a^2-u^2}$.

5. Suppose that the trig substitution $x = \sec(\theta)$ is being used on an integral where $x < 0$.
- a) How does the fact that $x < 0$ allow you to change that angle restriction that would need to put on θ ? Determine the smallest angle restriction that can be used for θ .
- b) Based on the angle restriction that you established in part (a) determine whether or not the statement $\theta = \sec^{-1}(x)$ is true or false.
- c) If $\pi \leq \theta < \frac{3\pi}{2}$ and $x < 0$, then the statement $\theta = C - \sec^{-1}(x)$ is always true (where C is some specific constant). Determine the value of C which makes this statement true.

6. Determine the integral $\int \frac{x^2 - 1}{x\sqrt{x^2 - 4}} dx$ assuming that $x < 0$.